Graduate Admissions: LIS 4761

Final Project 3: Group 16

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Overview

- Dataset
- Project Goal
- Methods Used
- Key Results
- Practical Implications
- Shiny App Demo
- Conclusion

About the dataset

- We used the Graduate Admissions dataset from Kaggle. It has 500 applicants and 8 variables.
- Source: https://www.kaggle.com/datasets/mukeshmanral/graduates-admission-prediction
- Dataset File: https://drive.google.com/file/d/1lPBzxyuVdM4XerKpWRI8zKaVerbP1B9Q/view?usp=drive_link
- The dataset includes GRE scores, TOEFL scores, CGPA, university rating, SOP and LOR strength, research experience, and the final chance of admission.
- Includes academic and personal factors, letting us compare which areas play a bigger role in predicting admission chances.

Cleaning the Data

- Imported raw CSV into R
- Standardized variable names
- Removed duplicate rows
- Trimmed extra spaces in text fields
- Removed incomplete observations
- Verified no missing values in *chance_of_admit*
- Saved final dataset as **admissions_clean.csv** for group modeling and visualization

Cleaned Dataset Preview

- These are a few of the main variables from the cleaned dataset

gre_score	toefl_score	university_rating	sop	lor	cgpa	research	chance_of_admit
337	118	4	4.5	4.5	9.65	1	0.92
324	107	4	4.0	4.5	8.87	1	0.76
316	104	3	3.0	3.5	8.00	1	0.72
322	110	3	3.5	2.5	8.67	1	0.80
314	103	2	2.0	3.0	8.21	0	0.65
330	115	5	4.5	3.0	9.34	1	0.90
321	109	3	3.0	4.0	8.20	1	0.75
308	101	2	3.0	4.0	7.90	0	0.68
302	102	1	2.0	1.5	8.00	0	0.50
323	108	3	3.5	3.0	8.60	0	0.45

Project Goal & Methods

- Project Goals
  - Identify which applicant factors most strongly influence admission likelihood
  - Compare how different analytics methods evaluate the same predictors
  - Understand interactions between academic, qualitative, and experiential strengths
  - Build interpretable and practical insights for real admissions decisions
  - Use and build upon what we learned from course lessons: Regression, Classification, and Visualization.
- Methods
  - Linear Regression
  - Logistic Regression
  - SVM Classification
  - Visualization – Shiny App

Linear Regression

- Used because *chance_of_admit* is a continuous numerical outcome
- Evaluates all predictors simultaneously to measure combined effects
  - Includes: GRE, TOEFL, University Rating, SOP, LOR, CGPA, Research
- Produces coefficient estimates, significance levels, and overall model fit
- Helps determine which variables contribute most after controlling for others
- Strong association with CGPA, followed by GRE, TOEFL, LOR, and research.
- University Rating and SOP showed a weak relationship
- Model explains **82% of variance**, indicating strong predictive structure

Logistic Regression

- Converted *chance_of_admit* into high vs. low likelihood groups using the median
- Logistic regression chosen for interpretable, odds-based classification
- Uses the same predictors as linear regression for consistency
- Estimates how each variable affects the probability of being “high likelihood”
- Evaluated with a confusion matrix to assess classification performance
- CGPA, GRE, TOEFL, and research are the biggest contributors
- Model achieved **87% accuracy**, serving as our baseline classifier

Support Vector Machine (SVM)

- Trained on the same binary admission labels as logistic regression
- Selected because SVM can capture nonlinear or boundary-based patterns
- Uses the same full predictor set to keep model comparisons aligned
- Constructs an optimal separating hyperplane between admission groups
- Evaluated using confusion matrix metrics on the test set
- Achieved **85% accuracy**, allowing comparison to logistic regression performance

Key Findings and Patterns

- CGPA is the strongest and most consistent predictor
  - Dominates in linear regression, logistic regression, and SVM
  - Suggests long-term academic performance outweighs test scores
  - Stronger effect than expected based on simple correlations
- SOP & LOR matter, but only in subtle ways
  - LOR significant in linear regression; SOP is weak
  - Both lose impact in logistic regression and SVM
  - Suggests they refine predictions (small adjustments) rather than separate high vs. low likelihood applicants
- University Rating shows nonlinear behavior
  - Weak in linear models
  - Still improves SVM classification
  - Indicates threshold effects (e.g., rating 4→5 matters more than 2→3)

Key Findings and Patterns

- Research helps, but not as much as assumed
  - Statistically significant, small effect size
  - Helps strong applicants but doesn’t compensate for low academics
  - Does not strongly influence behavior in SVM or logistic classification
- GRE and TOEFL influence different parts of prediction
  - GRE affects precise admission score (linear regression)
  - TOEFL better distinguishes borderline cases (logistic regression)
  - Shows exams are complementary, not interchangeable
- Predictive patterns are mostly linear
  - Logistic accuracy: 87%
  - SVM accuracy: 85%
  - Small performance gap suggests admissions patterns follow mostly linear boundaries

Why this is Relevant

- **Helps applicants understand what truly influences admission.**

Our results highlight that long-term academic performance plays a far larger role than many students expect. This gives prospective applicants a clearer sense of where their efforts will have the greatest impact.

- **Provides actionable guidance for advisors and mentors.**

Instead of vague encouragement, advisors can say things like, “Your SOP is strong, but a small increase in GRE could shift you into a higher-likelihood group,” supported by real data.

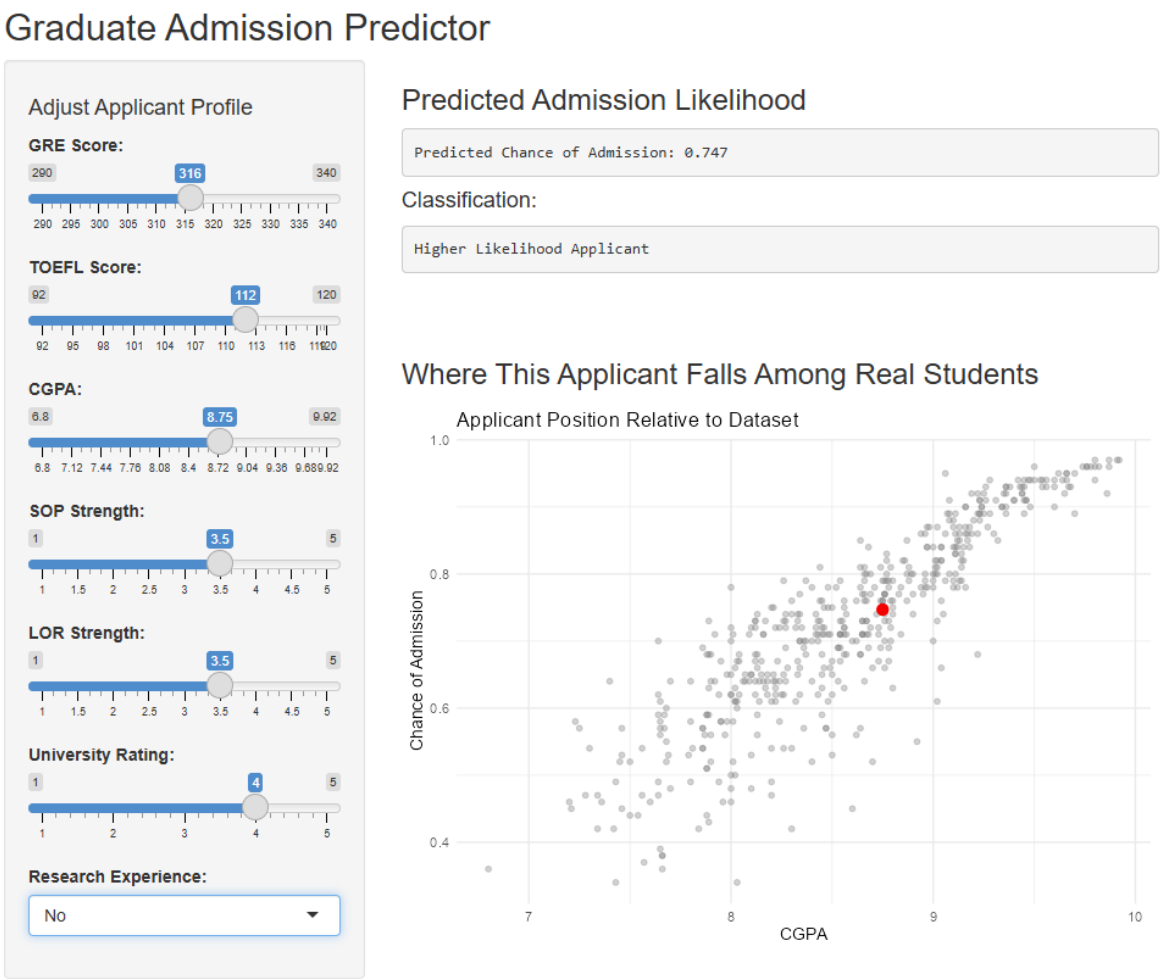
- **Encourages schools to reflect on their admissions balance.**

If academic variables outweigh qualitative components more than intended, institutions may reconsider how they balance their evaluation criteria to ensure the process aligns with their goals and values.

Visualization – Interactive Shiny App

Explore Admission Predictions

- Allows users to adjust applicant inputs (CGPA, GRE, TOEFL, SOP, LOR, University Rating, Research Experience).
- App instantly updates the predicted probability using our regression model.
- Visualizes the prediction on top of the full dataset for context.
- Demonstrates how different applicant profiles shift the likelihood of admission



Conclusion

- Long-term academic performance, especially CGPA, has the strongest influence, while GRE, TOEFL, and research experience each contribute in different but meaningful ways.
- This knowledge is helpful for applicants, advisors, and institutions.
- Understanding which variables truly drive admission helps correct common misconceptions and gives applicants clearer guidance on where to focus their effort.
- Taken together, our results provide practical guidance for students preparing applications and offer admissions committees a data-driven perspective on how their criteria function in practice.
- This project helped us build stronger analytical skills and gave us clearer insight into how to approach future modeling work.